RICK MILLER SME STANDARD

CANADA 2026 MINING HEAVY CIVIL ESTIMATING MANUAL & RATE LIBRARY

Consolidated Schedule of Rates, Bid Support Package & Estimating Standard
AACE 18R-97 Class 3 Baseline | Canada-wide Q2-2026 | All figures CAD

Field	Value
DOCUMENT NUMBER	RMM-CIVIL-CANADA-2026-MANUAL-REV3
REVISION	Rev 3 — End-to-End + Build-Up Decomposition + Restored Standby/Bonding Items
PREPARED BY	Rick Miller, Senior Project Director / EPCM
ISSUE DATE	Q2 2026
BASE CURRENCY	Canadian Dollar (CAD)
ESTIMATE CLASS BASIS	AACE 18R-97 Class 3 (±20% / +30%)
ESCALATION BASIS	Ausenco go-by Q2-2020 × 1.25 BCPI → Q2-2026
SOURCE DATA	RMM-CIVIL-CONTRACTOR-CANADA-2026-REV0 + RMM-CIVIL-CANADA-2026-REV0
INTENDED USERS	Contractor estimators, EPCM cost engineers, owner cost teams
CONFIDENTIALITY	Internal — not for public quotation issue without project validation

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1. EXECUTIVE SUMMARY

1.1 Purpose

This document is the consolidated, internally-maintained estimating standard and rate library for Canadian mining heavy civil construction. It is designed to be used by contractors, EPCMs, and owner cost teams across the full project lifecycle: budgetary pricing, feasibility-class estimates, tender support, bid evaluation, change pricing, and unit-rate validation.

1.2 Baseline Content (Extracted from Source Workbooks)

- 12 crew codes (B1–N1) — labour + equipment all-in $/hr with 2020 → 2026 escalation
- 27 fully-burdened labour positions covering supervision, civil trades, blasting, mechanical, electrical, instrumentation, PM and HSE
- 30 fully-operated equipment items with fuel burn, mob/demob flags, and effective availability
- ~78 Heavy Civil unit rates across 8 sections (earthworks → closure)
- ~22 contractor-scope unit rates (blasting, dewatering, buried services, paving, fencing, signage, rock support, culverts, crushing)
- 29 bid-screening floors/ceilings for tender evaluation
- 19 bonding & insurance line items including WCB rates by province
- 25 contractor overhead build-up items
- 19 standby & schedule items
- SME factors: design growth (Class 4 → 1), wastage, 10 Canadian regional zones, escalation, AACE accuracy ranges

1.3 Framework Templates (Authored in This Manual)

The following sections are industry-standard scaffolding sections, clearly marked as [TEMPLATE] in the corresponding Excel tabs. They are not drawn from the source workbooks — they are blank or sample structures to be populated per project:

- Equipment spreads by scope (Section 7)
- Unit rate build-up calculator (Section 9)
- Constructability and execution assumptions (Section 17)
- Scope inclusions, exclusions and clarifications register (Section 18)
- Bid leveling tender comparison (Section 19)
- Change order pricing methodologies (Section 20)
- Project estimate roll-up sheet (Section 21)
- Risk and contingency register (Section 22)

1.4 Estimate Class Applicability

AACE Class	Project Stage	Application in This Manual
Class 5	Concept screening / ROM	Use floor/ceiling ranges; apply +15-20% design growth
Class 4	Pre-feasibility / study	Mostly stochastic; equipment-factored; +15-20% DG
Class 3	Feasibility / budget (baseline)	Mid-range unit rates; +10-15% DG
Class 2	Sanction / AFE / pre-FID	Use mid-low range; +5-10% DG; firm quotes long-leads
Class 1	Tender / check estimate	Project-specific RFQ; low end of range; +2-5% DG

1.5 Key 2026 Benchmarks (mid-range)

- Mass earthworks rock (blasted): $55–$120 / m³

- Engineered fill compacted: \$22–\$45 / m^3
- Haul road full build (sub-grade to surface): \$65–\$120 / m^2
- TMF embankment Zone A/B: \$22–\$45 / m^3
- HDPE 2 mm liner (full S+I+CQA): \$28–\$55 / m^2
- CSP culvert 1200 mm: \$1,000–\$1,600 / m
- Concrete wall / grade beam: \$2,500–\$3,200 / m^3
- Rebar Grade 400 (S+I): \$4,000–\$6,500 / tonne — CBSA 25% surtax exposure
- Performance bond medium contract (\$5–\$25M): 1.0–1.8% of contract value
- Regional adjustor range: 1.00 (MB/SK baseline) → 1.55 (Yukon/NWT/Nunavut)
- Winter construction premium: 10–25% (all Canada)

1.6 Boundaries of Use

This manual provides a defensible baseline for internal estimating and bid evaluation. It is not a public quotation. Do not issue any unit rate as fixed pricing without project-specific validation, including supplier RFQs for long-lead and bulk materials, confirmation of WCB classification by province, verification of CBSA surtax exposure on steel and rebar, and site-specific productivity assumptions.

2. DOCUMENT CONTROL, BASIS & SOURCE DATA PROVENANCE

2.1 Source Workbooks

File Code	Title	Content
RMM-CIVIL-CANADA-2026-REV0	EPCM Benchmark Estimator (Rev E)	Crew Rates, Labour Rates, Equipment Rates, Heavy Civil Unit Rates, SME Factors
RMM-CIVIL-CONTRACTOR-CANADA-2026-REV0	Contractor Bid Tool (Rev G)	Bid Summary, Additional Unit Rates, Bonding & Insurance, Overhead, Standby, Bid Screening

2.2 Currency, Base Period & Exchange

All figures are in Canadian Dollars (CAD), basis Q2-2026. USD/CAD reference: 1.36–1.37 (Bank of Canada, May 2026).

2.3 Escalation Basis

Unit rates inherited from a Q2-2020 go-by have been escalated by a single cumulative factor of x1.25, representing the BCPI midpoint to Q2-2026. Rates re-baselined to 2026 directly (per supplier quotes or current published WCB / fuel tables) are flagged in each tab's Source/Basis column.

2.4 Confidentiality & Use

This document is intended for internal estimating, bid support and benchmarking. Distribution beyond a project team should be limited and approved. Any external use must include a current price-validity statement and exclusions register (Section 18).

3. ESTIMATING STANDARDS & METHODOLOGY (AACE 18R-97)

3.1 Estimate Classification Framework

This manual adopts AACE International Recommended Practice 18R-97 as its classification framework. The 5-class scheme is interpreted for mining heavy civil work as follows:

Class	Stage	Def.	Methodology	Accuracy	Use
Class 5	Concept screening	0–2%	Stochastic / parametric	-50% / +100%	ROM
Class 4	Pre-feasibility	1–15%	Mostly stochastic; equip-factored	-30% / +50%	Concept evaluation
Class 3	Feasibility / budget	10–40 %	Semi-detailed; unit rates; MTOs	-20% / +30%	Budget (this baseline)
Class 2	Sanction / AFE	30–75 %	Detailed; firm long-lead quotes	-10% / +20%	FID approval
Class 1	Tender / check	65–100 %	Detailed; IFC drawings; RFQs	-5% / +10%	Tender bid

3.2 Design Growth Allowances

Class	Discipline	Low	High	Notes
Class 5 / 4	Earthworks / civil	15%	20%	Limited geotech; preliminary drawings
Class 3	Earthworks / civil (this baseline)	10%	15%	Standard level for feasibility / budget
Class 2	Earthworks / civil	5%	10%	Pre-FID; IFC packages near complete
Class 1	Earthworks / civil	2%	5%	Full detailed design completed
All classes	Concrete works (additive)	+5%	+5%	Add 5% incremental to earthworks DG

3.3 Wastage Factors

Material	Low	High	Basis
Granular aggregate	5%	10%	Rounding + delivery variance
Concrete (formed)	8%	12%	Pump waste + edge loss
Rebar	3%	7%	Cutting waste; depends on bar size and complexity
Geomembrane liner	5%	10%	Panel overlaps + anchor trench + QA sampling
Geotextile	10%	15%	ASTM D4439 — 6-inch overlap minimum

3.4 Escalation & Market Basis (Q2-2026)

Item	Low	High	Source
Diesel — NS (May 2026)	$2.13/L	$2.21/L	NS Energy Regulatory Board
Diesel — AB (May 2026)	$1.68/L	$1.78/L	NRCan
Diesel — BC Metro (May 2026)	$2.15/L	$2.35/L	NRCan
USD/CAD (May 2026)	1.36	1.37	Bank of Canada
CBSA surtax — Chinese structural steel	25%	25%	CBSA Notices 24-26 / 25-22
CBSA surtax — Chinese rebar	25%	25%	Confirm supplier origin certificate
Labour CBA escalation 2024–2026 (NL)	3.5%	5.5%	Per trade — see Section 5

4. CREW COMPOSITIONS & PRODUCTION ASSUMPTIONS

Twelve standard crew codes are defined. Crew rates are reported both at the 2020 go-by basis and at the 2026 escalated basis (x1.25). Productivity (Prod Factor) is applied at the unit rate level (Section 8).

Code	Description	Avg Size	2020 Total $/hr	2026 Total $/hr
B1	Civil Works — Light	26.1	$127.50	$159.37
B2	Civil Works — Medium	28.1	$158.90	$198.63
B3	Civil Works — Heavy	37.7	$161.30	$201.63
B4	Civil Liner & Pipeline	28.1	$116.10	$145.12
C1	Concrete	22.6	$109.40	$136.75
D1	Structural Steel	17.6	$114.10	$142.63
E1	Architectural	19.6	$100.30	$125.37
G1	Mechanical Services	17.7	$111.30	$139.13
H1	Mechanical Equipment	23.6	$115.80	$144.76
J1	Piping	13.7	$110.50	$138.12
K1	Electrical	10.8	$117.80	$147.25
N1	Instrumentation	10.4	$123.80	$154.75

Crew composition by trade is shown in Section 5. The crews above are SME averages for benchmarking. For tender or sanction-class estimates, build the project-specific crew from Section 5 labour rates and Section 6 equipment rates.

5. LABOUR RATES — CANADA 2026 (FULLY BURDENED)

Labour rates are fully burdened: base wage + statutory burdens + employer benefits + WCB (province-specific) + small tools allowance + supervision ratio. Camp/LOA (rows O2/O3) is shown as additive per-person cost.

Ref	Position	Class	2026 Low	2026 High
S1	General Superintendent	Supervisor	$154	$172
S2	Superintendent	Supervisor	$136	$155
S3	General Foreman	Supervisor	$125	$140
C1	Civil Work Lead	Civil	$116	$132
C2	Equip Op — Heavy Duty	Civil	$110	$128
C3	Equip Op — Medium Duty	Civil	$101	$116
C4	Equip Op — Light Duty	Civil	$93	$106
C5	Labourer — Journeyman	Civil	$82	$95
C6	Labourer — 2yr Apprentice	Civil	$73	$84
C7	Labourer — 1st yr Apprentice	Civil	$64	$74
D1	Driller — Journeyman	Civil/Blast	$93	$108
D2	Blaster — Journeyman	Civil/Blast	$93	$108
M1	Mechanic — Heavy Duty	Mechanical	$84	$98
M2	Mechanic — Medium Duty	Mechanical	$92	$106
T1	Carpenter — Journeyman	Trades	$109	$125
T3	Concrete Lead	Trades	$125	$140
T4	Concrete Finisher	Trades	$95	$108
T5	Ironworker (rebar/struct)	Trades	$115	$132
T6	Welder / Pipefitter	Trades	$113	$130
T8	Rodman — Reinforcement	Trades	$94	$108
E1	Electrician — Journeyman	Electrical	$127	$145
I1	Instrumentation Tech	Instrumentation	$127	$145
P1	Project Engineer — Field	PM/Eng	$109	$125
P2	Project Engineer — Senior	PM/Eng	$123	$140
O1	Safety Officer (NCSO)	HSE	$110	$125
O2	Camp — all-in per person/day	Indirect	$220	$300
O3	LOA subsistence — non-camp	Indirect	$180	$250

6. EQUIPMENT RATES — CANADA 2026 (FULLY OPERATED)

Equipment rates are fully operated. Fuel is L/hr; multiply by site diesel price for fuel-cost portion. Availability % drives effective hours per shift.

Cat	Equipment	Size	Low $/hr	High $/hr	Fuel	Mob	Util
HT	Tandem dump truck	18-20T	$140	$165	45-65	Required	85%
HT	Tri-axle dump truck	25T	$155	$180	55-75	Required	85%
HT	Wiggle wagon / Super B	45T	$185	$215	65-90	Required	80%
HT	Off-road haul truck	40-100T	$320	$500	80-140	Included	85%
EX	Hyd excavator Cat 320	20-25T	$250	$330	18-25	Required	85%
EX	Hyd excavator Cat 336	35-40T	$340	$430	25-35	Required	85%
EX	Hyd excavator Cat 390	50-65T	$460	$580	38-50	Required	82%
EX	Long-reach excavator	20T 18m	$290	$380	18-25	Required	82%
DO	Track dozer Cat D6N	17T	$250	$320	20-30	Required	88%
DO	Track dozer Cat D8T	37T	$380	$480	35-48	Required	88%
DO	Track dozer Cat D10T	58T	$520	$650	50-68	Required	85%
GR	Motor grader Cat 140M	-	$280	$360	18-26	Required	88%
GR	Motor grader Cat 16M	large	$360	$450	22-32	Required	88%
CP	Vib roller CS56B	12T	$200	$270	15-22	Required	88%
CP	Padfoot CS74B	17T	$240	$310	18-26	Required	88%
CP	Plate compactor	0.5-1T	$45	$75	5-8	By truck	90%
WL	Wheel loader Cat 930M	10T	$200	$270	18-26	Required	87%
WL	Wheel loader Cat 950M	16T	$250	$320	22-30	Required	87%
WL	Wheel loader Cat 980M	22T	$320	$400	28-38	Required	87%
DR	Rotary drill rig	120mm hole	$550	$750	35-50	Included	80%
SK	Skid steer Cat 262D	-	$95	$135	10-15	By truck	90%
WP	Diesel submersible pump	4-inch	$55	$85	10-16	By truck	90%
WP	Diesel submersible pump	8-inch	$120	$180	20-30	By truck	90%
CR	RT crane Grove	RT 60T	$650	$900	30-45	Required	80%
CR	Crawler crane Liebherr	150T	$1,800	$2,600	60-90	Included	78%
MI	Rock breaker Cat 345+hammer	-	$480	$620	28-40	Required	80%
MI	Air compressor diesel	375 cfm	$75	$110	12-18	By truck	90%
MI	Light tower diesel	8-10 kW	$35	$55	5-8	By truck	95%
MI	Fuel tanker site supply	10,000L	$180	$250	30-40	Required	90%
MI	Water truck dust	10,000L	$160	$220	30-40	Required	90%

7. EQUIPMENT SPREADS BY SCOPE [TEMPLATE]

Typical SME spreads for Class 3 estimating. Validate fleet sizing against site access, haul distance, geotechnical conditions, climate window, and crew availability before committing to a bid.

Scope	Primary + Support Equipment	Crew	Production
Mass earthworks — rock	1× DR rotary, 1× EX Cat 390, 4–6× off-road 40-100T	B3	800–1,500 m³/shift
Mass earthworks — common	1× EX Cat 336, 4–6× tandem/tri-axle, 1× D8T	B3	1,200–2,000 m³/shift
Engineered fill placement	1× EX Cat 320 or WL 950M, 2–4× tandem, 1× CS74B	B2	600–1,000 m³/shift
Haul road build	1× D8T (sub-base), 1× 140M grader (surface)	B2	150–300 m/day
HDPE liner install (TMF)	1× WL 930M, 1× SK, welding crew	B4	1,500–3,000 m²/day
Culvert install (≥900 mm CSP)	1× EX Cat 320, 1× WL 930M, 2× tandem, dewatering	B4	20–40 m/day
Buried services trenching	1× EX Cat 320, 1× tandem, 1× WL 930M	B4	60–120 m/day
Concrete pour — foundation	1× crane RT 60T, 1× concrete pump, 1× vibrator	C1	12–25 m³/day
Rock support — shotcrete	1× shotcrete pump, 1× EX (mesh handling), 1× SK	C1	120–250 m²/day
Drill + blast rock anchors	1× drill rig, 1× EX, 1× grout plant	D1	8–15 anchors/day
Buried HDPE forcemain	1× EX Cat 320, 1× WL, 1× fusion machine, 1× tandem	B4	80–150 m/day
Demolition / strip and grub	1× D8T, 1× EX Cat 336, 1× tandem	B3	0.5–1.5 ha/day

8. MASTER SCHEDULE OF RATES

The full line-item Schedule of Rates is provided in the companion Excel workbook (sheet '08 SoR Master'). It contains ~110 line items with unit man-hours, productivity factor, labour / equipment / material / sub build-up columns, CDI %, design growth %, wastage %, and crew code per item. The PDF below summarises the section structure.

8.1 Section Structure

Section	Scope	Source	Items
A — Earthworks & Mass Excavation	Common/rock excavation, fill, granular, drill+blast (holes only)	Rev E	15
B — Water Management / SWM & ESC	Riprap, geotextile, silt fence, temp dewatering	Rev E	10
C — Roads, Pads & Working Surfaces	Sub-grade, granular base, asphalt, equip pads	Rev E	8
D — WRSA / TMF Embankment	Embankment fill, HDPE liner, drainage layers, riprap ditching	Rev E	8
E — Drainage, Culverts & Piping	CSP, HDPE, flow guards, discharge piping	Rev E	15
F — Concrete	Walls, piers, footings, shotcrete, rebar	Rev E	5
G — Structural Steel	Beams, columns, grating, handrail, stairs, deck, cladding	Rev E	8
H — Closure & Reclamation	Cap, cover, topsoil, HDPE cap liner	Rev E	4
I — Blasting (full scope)	Drill, charge, blast, clear, controlled / presplit	Rev G	4
J — Permanent Dewatering	Pump stations, forcemains, wellpoint systems	Rev G	5
K — Buried Services	Water, sewer, manholes, thrust blocks, valves, hydrants	Rev G	7
L — Concrete Paving & Curbing	Paving, curb & gutter, wheel wash	Rev G	3
M — Fencing, Security & Wildlife	Chain link, wildlife exclusion electrified, gates	Rev G	4
N — Site Signage & Traffic Mgmt	Flagging, dust suppression	Rev G	2
O — Rock Anchors & Stabilization	Rock anchors, wire mesh, shotcrete + mesh	Rev G	4
P — Culvert Structures	Precast box, arch, headwalls	Rev G	3
Q — Aggregate Crushing On-Site	Granular A/B, riprap — crushing operating cost	Rev G	2
R — Underground Civil [Rev 2]	Shaft sinking, lateral dev, raise bore, ground support, mucking	Rev 2	12
S — Tailings Dam Zoned Construction [Rev 2]	Zoned starter dam + raises, internal drainage, instrumentation	Rev 2	13
T — Heap Leach Pad Construction [Rev 2]	Multi-liner system, drain rock, manifolds, PLS pond	Rev 2	10
U — Process Plant Civil [Rev 2]	Mill / crusher / thickener / flotation foundations, conveyor structures	Rev 2	11
V — Slurry / Concentrate / Fuel Pipeline [Rev 2]	ROW, trenching, HDPE/steel pipe, valve stations, HDD	Rev 2	10
W — Mine Site Bridges & Crossings [Rev 2]	Modular, concrete, haul road bridges, abutments, large culverts	Rev 2	6
X — Power Line & Substation Civil [Rev 2]	Tower foundations, distribution poles, substation pads	Rev 2	7
Y — Hydraulic Structures [Rev 2]	Diversion channels, riprap classes, dissipators, spillways	Rev 2	9

Rev 2 totals: 185 line items across 25 sections (A–Y). Specialty mining sections (R–Y) added in Rev 2 from SME judgement; ranges are Class 3 SME baselines pending project-specific RFQ confirmation.

9. UNIT RATE BUILD-UP METHODOLOGY

The standard SME unit-rate composition is a six-element build-up:

Element	Symbol	Source	Application
Unit Labour	L	Crew rate × (Unit MH / Productivity)	Section 4–5
Unit Equipment	E	Same crew driver — included in crew total	Section 4 / 6
Unit Material	M	Supplier quote × (1 + Wastage)	Section 3.3 / supplier RFQ
Unit Subcontract	S	Sub quote (bare, before markup)	Per sub
Contractor Indirect	I	% of (L+E+M+S) — CDI overhead	Section 11 / 14
Total direct unit rate	U	= (L + E + M + S) × (1 + I)	Sheet 09 calculator

Two further factors are applied at unit-rate level before roll-up:

Factor	Symbol	Range	Notes
Design growth %	DG	2–20% depending on AACE class	Apply to quantity, not to unit rate
Regional adjustor	R	1.00 – 1.55	Apply to direct unit rate × (1+DG)

Final unit rate = U × (1 + DG) × R. The 6-layer markup stack (Section 15) is then applied at project roll-up, not at line level. The companion Excel workbook (sheet '09 Unit Rate Calculator') provides a live calculator with formulas.

10. BID SCREENING — FLOORS, CEILINGS & SENSE-CHECKS

Any bid line item priced more than 20% below the floor or 20% above the ceiling warrants a clarification request. Items below the floor most commonly indicate missing scope (insulation, CQA, supply, etc.).

Category	Benchmark Item	UOM	Floor	Ceiling
EARTHWORKS	Mass earthworks — rock (blasted)	$/m³	$55	$120
EARTHWORKS	Engineered fill compacted Zone A/B	$/m³	$22	$45
EARTHWORKS	Granular road base 150mm	$/m²	$22	$42
EARTHWORKS	Haul road full build	$/m²	$65	$120
EARTHWORKS	Strip and grub per hectare	$/ha	$8,500	$18,000
WRSA / TMF	TMF embankment engineered fill	$/m³	$22	$45
WRSA / TMF	HDPE liner 2mm supply+install	$/m²	$28	$55
WRSA / TMF	TMF perimeter ditch riprap lined	$/m	$280	$500
DRAINAGE	CSP culvert 1200mm supply+install	$/m	$1,000	$1,600
DRAINAGE	HDPE 150mm above grade insulated	$/m	$230	$380
DRAINAGE	Flow guard 600mm supply+install	$/ea	$2,200	$3,500
DRAINAGE	Flow guard 1200mm supply+install	$/ea	$8,000	$12,000
CONCRETE	Concrete foundation/wall	$/m³	$2,200	$3,500
CONCRETE	Shotcrete with fibre 75mm	$/m²	$90	$160
CONCRETE	Rebar supply+install	$/tonne	$3,800	$6,500
BLASTING	Pre-blast survey per structure	$/struct	$3,000	$9,000
FENCING	Wildlife exclusion electrified 2.4m	$/m	$150	$280
BURIED SERVICES	Sanitary manhole precast 1200mm	$/ea	$4,000	$8,000
OVERHEAD CHECK	Contractor OH+profit — $50M project	% dir.	14%	22%
OVERHEAD CHECK	Performance + L&M; bond (>$25M)	% cont.	0.75%	1.25%

11. CONTRACTOR INDIRECTS (CDI) & OVERHEAD BUILD-UP

Typical project total: 12–18% of direct cost. Detail below from Rev G source workbook.

Category	Item	UOM	2026 Range (CAD)
FIELD OFFICE	Site trailer — lunchroom/dryroom	month	$1,500–$2,500
FIELD OFFICE	Temp power — 100 kW diesel generator	month	$4,500–$8,500
FIELD OFFICE	Temp power — 250 kW generator large camp	month	$9,500–$16,000
FIELD OFFICE	Site IT, internet, comms (Starlink)	month	$850–$2,500
FIELD OFFICE	Site vehicle — project pickup truck	month	$1,200–$1,800
FIELD OFFICE	Site vehicle — 20-seat crew bus	month	$3,500–$6,500
FIELD OFFICE	Mob/demob — site setup + teardown	LS	$35,000–$85,000
PM STAFF	Senior Project Engineer	month	$14,000–$20,000
PM STAFF	Field / QC Engineer	month	$11,000–$16,000
PM STAFF	Site Safety Officer (NCSO certified)	month	$12,000–$17,000
PM STAFF	Senior Superintendent	month	$16,000–$24,000
PM STAFF	General Foreman	month	$13,000–$19,000
PM STAFF	Document Controller / Field Admin	month	$7,500–$11,000
QC & TESTING	Nuclear density testing — crew/day	day	$850–$1,500
QC & TESTING	Concrete cylinder testing — per set	set	$185–$350
QC & TESTING	Aggregate gradation testing	test	$250–$450
QC & TESTING	Geomembrane weld testing (spark test)	m weld	$2.50–$4.50
QC & TESTING	Survey control — survey crew/day	day	$2,500–$4,500
ENVIRONMENTAL	Spill kit and containment supplies	month	$350–$750
ENVIRONMENTAL	Environmental water sampling	sample	$450–$950
ENVIRONMENTAL	Environmental coordinator (part-time)	month	$5,500–$9,500
HEAD OFFICE	Warranty / deficiency holdback allowance	% cont.	0.5%–1.0%

12. BONDING & INSURANCE

WCB is province-specific — confirm classification and base rate against the relevant authority before pricing.

Category	Item	Basis	Rate (CAD)
SURETY	Performance Bond — small (<$5M)	% of contract	1.5%–2.5%
SURETY	Performance Bond — medium ($5–$25M)	% of contract	1.0%–1.8%
SURETY	Performance Bond — large (>$25M)	% of contract	0.75%–1.25%
SURETY	Labour & Material Payment Bond	% of contract	0.5%–1.0%
SURETY	Maintenance/Warranty Bond — 12 month	% of contract	0.5%–1.0%
LIABILITY	Contractor's Pollution Liability	Annual policy	$1,500–$3,500
WCB	Labourer — Alberta WCB Industry 41	Per $100 payroll	$3.80–$5.20
WCB	Labourer — BC WorkSafeBC Class 711001	Per $100 payroll	$4.50–$6.00
WCB	Labourer — NS WCB NS	Per $100 payroll	$3.80–$5.50
WCB	Labourer — NL WorkplaceNL	Per $100 payroll	$4.00–$5.80
WCB	Labourer — Quebec CNESST	Per $100 payroll	$5.50–$8.00
WCB	Equipment Operator — all provinces	Per $100 payroll	$3.50–$5.00
WCB	Blaster / Driller — elevated risk	Per $100 payroll	$5.50–$8.00
PROPERTY	Contractor's Equipment Floater	% of fleet repl/yr	0.5%–1.2%

13. STANDBY RATES, ACCELERATION & DELAY

Category	Item	UOM	2026 Range (CAD)
EQUIP STANDBY	Excavator Cat 320 — standby	hr	\$110–\$155
EQUIP STANDBY	Excavator Cat 336 — standby	hr	\$140–\$200
EQUIP STANDBY	Track dozer D8T — standby	hr	\$155–\$215
EQUIP STANDBY	Motor grader Cat 140M — standby	hr	\$110–\$155
EQUIP STANDBY	RT crane 60T — standby	hr	\$320–\$480 (4hr min)
EQUIP STANDBY	Crawler crane 150T — standby	hr	\$850–\$1,300
LABOUR STANDBY	Equipment operator — standby	hr	\$93–\$128
LABOUR STANDBY	B3 crew of 10 — standby	day	\$7,000–\$10,000
ACCELERATION	Overtime — double time (Sun/holiday)	% add	100% / 100%
ACCELERATION	Afternoon shift premium	% add	5%–10%
ACCELERATION	Night shift premium	% add	10%–15%
WINTER	Cold weather heating — LP heaters	day	\$450–\$850
WINTER	Cold weather enclosure — concrete	m² enc.	\$8–\$18
WINTER	Winter freeze protection — earthworks	m³ thaw	\$3.50–\$8.50
OWNER DELAY	Equip demob + remob per event	event	\$15,000–\$65,000
OWNER DELAY	Extended project overhead	week	\$12,000–\$35,000
OWNER DELAY	Material restocking / escalation impact	% mat	2%–8% (>3 mo)

14. SME FACTORS — REGIONAL ADJUSTORS & MARKUPS

14.1 Regional Adjustors (Canada 2026)

Apply the regional adjustor to the direct unit rate (after design growth). Baseline is MB/SK = 1.00–1.08.

Region	Low	High
Ontario (Greater Toronto / Ottawa)	1.05	1.15
Ontario (Northern — Thunder Bay / Sudbury)	1.02	1.10
Quebec (Montreal)	1.03	1.12
Quebec (Northern — Chibougamau / James Bay)	1.08	1.22
Manitoba / Saskatchewan	1.00	1.08
Alberta (Edmonton / Calgary)	1.05	1.18
Alberta (Oil Sands / Fort McMurray)	1.15	1.35
British Columbia (Metro Vancouver)	1.10	1.20
British Columbia (Interior / Northern)	1.15	1.30
Yukon / NWT / Nunavut	1.30	1.55
Winter construction premium — all Canada	10%	25%

14.2 Indirect / GC Markup Reference

Item	Low	High	Notes
Mob/demob major earthwork fleet (% of equip)	2%	5%	Or flat LS per fleet assembly
Mob/demob RT crane 60T	$4,000	$8,000	Flat rate per mob
Mob/demob crawler crane 150T	$25,000	$55,000	Incl. partial disassembly
Camp/LOA markup on direct labour	8%	12%	Add for remote sites
Small tools and consumables (on direct lab.)	2.5%	4%	Blades, hoses, bits, PPE
Supervision markup on direct labour	8%	12%	Foreman + super ratio
Contractor distributable (CDI) on direct cost	6%	10%	Temp facilities, insurance, site admin

15. CONTRACTOR BID COST STRUCTURE — 6-LAYER MARKUP MODEL

All contractor bids in this manual are composed of six stacked cost layers. The first layer (Direct Field Costs) is built up from the Schedule of Rates (Section 8); the remaining five layers apply % markups in the following sequence:

Layer	Scope	Low (%)	High (%)
1 — Direct Field Costs	Unit rates × MTO quantities	—	100% (basis)
2 — Contractor Field Indirects (CDI)	Temp facilities, supervision, small tools, QC, environmental	12%	18%
3 — Company Overhead	Bid & proposal, warranty, head office allocation	6%	12%
4 — Bonding & Insurance	Performance bond, L&M; bond, CGL, WCB, Builder's Risk	2%	4%
5 — Contingency & Risk	Ground conditions, weather, escalation, sub default	8%	15%
6 — Contractor Profit	Profit / fee — varies by risk and competitive position	8%	15%

Total bid = Direct + CDI + OH + Bond/Ins + Contingency + Profit. For mid-size mining civil projects (~$50M direct), total OH + profit typically falls in the 14–22% range — values outside this band warrant review.

16. MOBILIZATION & DEMOBILIZATION LOGIC

Mob/demob is typically priced as a separate lump-sum line, not absorbed in unit rates. For remote/fly-in sites, add freight allowance. For multi-phase work, charge re-mob per phase.

Category	Asset / Activity	Basis	2026 Range (CAD)
Earthwork fleet	Major earthwork fleet mob/demob	% of equip value or flat LS	2%–5%
Earthwork fleet	Heavy haul move per low-bed trip	Per round-trip	$1,800–$3,500
Earthwork fleet	Off-road haul truck (40–100T) move	Per unit / round-trip	$8,000–$18,000
Cranes	Rough terrain crane RT 60T	Flat per mob	$4,000–$8,000
Cranes	Lattice boom crawler 150T	Flat per mob (incl. disassembly)	$25,000–$55,000
Camps	Camp setup — fly-in remote (40-bed)	LS	$250,000–$500,000
Camps	Camp setup — drive-in (40-bed)	LS	$120,000–$220,000
Specialty subs	HDPE liner sub (incl. CQA crew)	LS	$25,000–$60,000
Specialty subs	Blasting sub (drill + powder magazine)	LS	$35,000–$90,000
Specialty subs	Aggregate crushing plant	LS	$150,000–$400,000
Insurance / WCB	Provincial registration + WCB account	LS	$5,000–$15,000

17. CONSTRUCTABILITY & EXECUTION ASSUMPTIONS [TEMPLATE]

This section captures the standing assumptions that frame each estimate. Populate per project — the headings below are mandatory; the example detail is illustrative.

17.1 Work Calendar

- Site works 10-hr shift, 14-on / 7-off (FIFO) OR 4-on / 3-off (drive-in)
- Effective hours per shift after breaks & travel: 8.5–9.0 productive hr
- Winter weather window (no work): Nov–Mar productivity factor 0.70–0.85
- Spring break-up moratorium (haul road restrictions): early Apr to mid-May

17.2 Access & Logistics

- Primary access road condition (paved / gravel / seasonal) and weight rating
- Fly-in vs drive-in — confirm chartered air freight cost & frequency
- Material laydown sufficient for 60–90 days inventory at site
- Fuel resupply schedule — confirm tanker turnaround and on-site storage capacity

17.3 Sequencing

- Earthworks completion before liner install (no traffic post-liner)
- Concrete cure times: 7 days form-strip; 28 days full strength; cold weather double
- Steel erection cannot proceed until anchor bolts surveyed and accepted
- Culvert installs during low-flow window per provincial fisheries authorization

17.4 Geotechnical

- Bedrock encountered at depth X m — drill+blast assumption from elevation Y
- Water table at depth X m — dewatering required for excavations deeper than Y m
- Acid-generating waste rock requires segregation and encapsulation

17.5 Environmental

- Fish habitat compensation work in Year 1 — site-specific permits required
- SWPPP / ESCP plan controlling discharge to receiving water; weekly inspection minimum
- Wildlife protocol (bear / caribou / grizzly) — daily site clearance

17.6 Construction Methods

- Roll-on / roll-off compaction at 95% Std Proctor; 98% for process pads
- Concrete supplied from on-site batch plant (if vol >2,000 m³) else ready-mix
- Aggregate supplied from on-site quarry/crusher (if vol >50,000 t) else from X km
- Blasting confined per pattern; pre-blast survey for structures within 500m radius

17.7 Interface

- Owner-supplied items (e.g. process plant equipment)
- Existing facility tie-ins coordinated during scheduled outage windows

17.8 Labour / IR

- Open shop / CLAC / Building Trades — confirm market and CBA obligations

- Indigenous community engagement commitment — % local hire / IBA obligations

17.9 HSE

- Site induction 4 hr per worker; site-specific safety standards apply
- Fire watch protocol for hot work; permit-to-work system in operation
- Critical lifts >75% chart capacity require lift plan + engineer approval

17.10 Productivity

- Apply 0.85–0.95 productivity factor for first 4 weeks of project (learning curve)
- Confined working areas: derate output by 10–30% depending on access

17.11 Quality

- All concrete pours: nuclear density + 4-cyl set per pour
- All HDPE liner seams: 100% non-destructive + destructive sample 1/150 m

18. SCOPE INCLUSIONS, EXCLUSIONS & CLARIFICATIONS [TEMPLATE]

The following register defines a typical contractor scope position. Populate INCLUDED / EXCLUDED / CLARIFY per project. This register travels with the bid and is referenced in the contract documents.

Discipline	Item	Status
EARTHWORKS	Mass excavation common soil	INCLUDED
EARTHWORKS	Rock excavation including drill & blast	INCLUDED
EARTHWORKS	Removal of contaminated soil	EXCLUDED
EARTHWORKS	Temporary site dewatering	CLARIFY
WATER MGMT	Permanent stormwater pond construction	INCLUDED
WATER MGMT	Operating sediment treatment chemicals	EXCLUDED
ROADS	Haul road sub-base + surface course	INCLUDED
ROADS	Permanent asphalt paving	CLARIFY
WRSA/TMF	TMF embankment construction Zones A, B, C	INCLUDED
WRSA/TMF	HDPE 2mm liner with CQA	INCLUDED
WRSA/TMF	Dam Safety Officer (DSO) services	EXCLUDED
DRAINAGE	CSP culverts up to 1200mm diameter	INCLUDED
DRAINAGE	Precast box / arch culverts > 1500mm	CLARIFY
CONCRETE	Cast-in-place foundations, walls, footings	INCLUDED
CONCRETE	Precast concrete (process plant)	EXCLUDED
STEEL	Building enclosures + equipment platforms	INCLUDED
STEEL	Process plant structural steel	EXCLUDED
GEOSYNTHETICS	HDPE liner systems	INCLUDED
GEOSYNTHETICS	Geomembrane CQA Engineer of Record	EXCLUDED
BLASTING	Drill, blast, clear of rock excavation	INCLUDED
BLASTING	Pre-blast condition surveys	INCLUDED
BURIED SVCS	Site potable water, sewer, process water	INCLUDED
BURIED SVCS	Electrical / instrumentation underground	EXCLUDED
FENCING	Site perimeter chain link + wildlife excl.	INCLUDED
FENCING	Process plant area security fencing	EXCLUDED
CLOSURE	Final cover system + topsoil + seeding	INCLUDED
CLOSURE	Long-term water treatment infrastructure	EXCLUDED
INDIRECTS	Site offices, lunchrooms, dryrooms	INCLUDED
INDIRECTS	Camp accommodation (incl. food)	CLARIFY
INDIRECTS	Owner inspection / EPCM offices	EXCLUDED
PERMITS	Routine construction permits	INCLUDED
PERMITS	Environmental authorization (EA, water lic.)	EXCLUDED
PERMITS	Provincial blasting permits, magazine licenses	INCLUDED
WARRANTY	12-month warranty	INCLUDED
WARRANTY	Extended warranty beyond 12 months	EXCLUDED

19. BID LEVELING — TENDER COMPARISON FRAMEWORK [TEMPLATE]

Use a side-by-side worksheet (companion Excel sheet '19 Bid Leveling') to compare bidder pricing against the SME benchmark. Convert each bidder's Form of Tender into a common UOM. Flag any item more than ±20% from the benchmark.

19.1 Process

1) Common UOM normalization — strip lump sums, convert per-unit, align UOM.

2) Build the comparison table: Item | Qty | Bidder A | Bidder B | Bidder C | SME Benchmark.

3) Compute variance % vs benchmark for each bidder line.

4) Aggregate top 10 deviation items by absolute $ — these drive total bid difference.

5) Compile clarification / deviation log per bidder; request written response.

6) Re-level after clarifications: this is the 'normalized bid' that goes forward to commercial evaluation.

7) Sense-check: bidder total ÷ direct quantity ($/m³ aggregate) against Section 10 ranges.

19.2 Common Deviation Categories

Typical sources of bid deviation: missing inclusions, different productivity assumption, different supplier source (especially steel/rebar after CBSA), regional adjustor misapplication, unbalanced bidding (front-loaded mobilization), and exclusion of CQA, surveys, or testing.

20. CHANGE ORDER PRICING METHODOLOGIES

Five methodologies are recognized. Methodology used should be agreed with the Contract Administrator at project start.

Method	When to Use	Build-Up
Unit-rate based	Quantum well-defined; existing rate covers scope	Qty × rate × (1+DG)
Time & Materials	Scope uncertain or schedule-critical	LEM hr × $/hr + Mat × (1+wastage) + Sub × (1+markup) + OH%
Lump-sum quotation	Discrete, well-defined; firm quote possible	Direct + Indirect 10–15% + OH+P 10–20%
Negotiated rates	Recurring change category	Pre-agreed rate schedule × Qty
Force account	Disputed scope; owner directs work pending resolution	T&M as above + auditable records

20.1 Standard Markup Stack (CCDC 2 GC 6.2.4)

Layer	Element	% Range	Apply To
1	Direct labour + equip + material	100%	Cost of work
2	Contractor field overhead	8–12%	Direct labour
3	Head office overhead	5–10%	Total direct + CFO
4	Contractor profit / fee	8–15%	Total direct + OH
5	Bond + insurance adjustment	1.5–3%	Total contract increase
6	Sub-tier markup (flow-through subs)	5–10%	Sub price

Daily T&M; sheets must be co-signed within 48 hours. Lump-sum change quotations must include itemised breakdown. Time-impact analysis is required if the change affects critical path (CCDC 2 GC 6.5).

21. PROJECT ESTIMATE ROLL-UP TEMPLATE [TEMPLATE]

Roll-up format mirroring the 6-layer bid model. Populate the Direct field cost lines from MTO × unit rate (Section 8). Apply markups within the bands shown in Section 15. Apply the regional adjustor (Section 14) to direct lines before summing if outside MB/SK baseline. Live formulas are in the companion Excel sheet '21 Project Roll-Up'.

Layer	Formula
1 — DIRECT FIELD COSTS	Σ Earthworks + WaterMgmt + Roads + WRSA + Drainage + Concrete + Steel + Geosynth + Blast + Buried + Fencing + Dewater + Closure
2 — CONTRACTOR FIELD INDIRECTS	= Direct × CDI% (typ. 12-18%)
3 — COMPANY OVERHEAD	= Direct × OH% (typ. 6-12%)
4 — BONDING & INSURANCE	= (Direct + CDI + OH) × B+I% (typ. 2-4%)
5 — CONTINGENCY & RISK	= Direct × Cont.% (typ. 8-15%)
6 — CONTRACTOR PROFIT	= (Direct + CDI + OH + B+I + Cont.) × Profit% (typ. 8-15%)
TOTAL BID	= Σ of layers 1-6

22. RISK & CONTINGENCY REGISTER [TEMPLATE]

Typical risk categories for Canadian mining heavy civil with illustrative likelihood / impact ranking. Total cost-weighted exposure drives the contingency carry in Section 15 Layer 5.

ID	Cat.	Description	L	I	Mitigation / Allocation
R-01	Geotechnical	Unforeseen rock below estimated quantity	4	4	3-8% earthworks contingency
R-02	Geotechnical	Acid-generating waste rock encountered	3	5	Owner-led env mgmt; 1-3% allowance
R-03	Weather	Extended winter shutdown beyond plan	4	3	2-5% weather contingency
R-04	Market	Diesel >15% above bid basis	3	3	3-8% materials escalation; fuel surcharge
R-05	Tariff	CBSA 25% surtax extended on steel/rebar	3	3	Source Canadian/US; 5-10% on steel
R-06	Subcontractor	Specialty sub default or delay	2	4	1-3% sub default contingency
R-07	Permits	Environmental permit delay	3	4	Phase work; engage authority early
R-08	Labour	Skilled trades shortage	4	3	Lock supply via early subs; 5% premium
R-09	Indigenous	IBA / community engagement delay	3	4	Engage early
R-10	Owner	Owner-supplied materials late	3	4	Daily T&M; / standby
R-11	Quality	HDPE liner CQA failures	2	3	Pre-qualify installer; third-party CQA
R-12	HSE	Major incident causing stop-work	1	5	Carry incident allowance

23. GLOSSARY & ABBREVIATIONS

Term	Definition
AACE	Association for the Advancement of Cost Engineering International
AACE 18R-97	Cost Estimate Classification System — process industries
AFE	Authorization for Expenditure — owner approval at sanction
BCPI	Building Construction Price Index (Statistics Canada)
CAD	Canadian Dollar
CBSA	Canada Border Services Agency
CDI	Contractor Distributable Indirects (site indirects)
CGL	Commercial General Liability insurance
CLAC	Christian Labour Association of Canada (open shop)
CCDC	Canadian Construction Documents Committee (CCDC 2 = stipulated sum)
CQA	Construction Quality Assurance (typically third-party)
DR-11 / DR-17	HDPE pipe Dimension Ratio (wall thickness class)
EHT	Employer Health Tax (Ontario / BC)
EPCM	Engineering, Procurement and Construction Management
ESC	Erosion & Sediment Control
FID	Final Investment Decision
FIFO	Fly-in / Fly-out (14-on / 7-off rotation)
HDPE	High-Density Polyethylene
HSE	Health, Safety, Environment
IBA	Impact Benefit Agreement (Indigenous communities)
IFC	Issued For Construction (drawing status)
LOA	Living Out Allowance (per-diem) for non-camp remote work
MTO	Material Take-Off
NCSO	National Construction Safety Officer (Canadian certification)
NRCan	Natural Resources Canada
ROM	Rough Order of Magnitude (Class 5/4 estimate)
SoR	Schedule of Rates
SWM	Stormwater Management
SWPPP	Stormwater Pollution Prevention Plan
TMF	Tailings Management Facility
WBS	Work Breakdown Structure
WCB / WSIB	Workers' Compensation Board / Ontario WSIB
WRSA	Waste Rock Storage Area
S+I	Supply + Install
S+F+P	Supply + Form + Pour

24. SOURCES & REFERENCES

Category	Reference
STANDARDS	AACE Recommended Practice 18R-97 — Cost Estimate Classification System
STANDARDS	ASTM D698 — Standard Proctor; ASTM D1557 — Modified Proctor
STANDARDS	ASTM D4439 — Geosynthetics terminology
STANDARDS	CCDC 2 — Stipulated Sum Contract
STANDARDS	Canadian Dam Association Dam Safety Guidelines
INDICES	Statistics Canada — Building Construction Price Index (BCPI), quarterly
INDICES	NRCan — Diesel Pricing, weekly
INDICES	Bank of Canada — USD/CAD exchange rate
AUTHORITIES	CBSA — Notices 24-26 and 25-22 (steel / rebar surtax)
AUTHORITIES	Provincial WCB / WSIB / CNESST / WorkSafeBC / WorkplaceNL
AUTHORITIES	NS Energy Regulatory Board — Atlantic diesel price
INTERNAL	Rev G workbook — RMM-CIVIL-CONTRACTOR-CANADA-2026-REV0 (Contractor Bid Tool)
INTERNAL	Rev E workbook — RMM-CIVIL-CANADA-2026-REV0 (EPCM Benchmark Estimator)
INTERNAL	Ausenco go-by Q2-2020 (Senior PD / EPCM reference)
SUPPLIERS	Armtec — CSP culvert and flow guard pricing (Dec-2025)
SUPPLIERS	Various HDPE pipe suppliers — Jan-2026 quotes +15% install factor
BENCHMARKS	SME benchmark observations 2024-2026 — Canadian mining civil contracts
METHODOLOGY	Standard industry practice — contractor markup stack and CDI percentages
METHODOLOGY	CCDC 2 GC 6.2 / 6.4 / 6.5 — change order provisions

Public-Tender Data Sources (Rev 2 — benchmark cross-reference)

Category	Reference
PUBLIC TENDER	Alberta Transportation — Unit Price Averages 2026 (XLSX) — weighted avg of 3 low bids, tenders May 1, 2024 – Sep 30, 2025
PUBLIC TENDER	Alberta Transportation — Unit Prices and Cost Adjustments — alberta.ca/unit-prices-and-cost-adjustments
PUBLIC TENDER	BC Road Builders — Tender Report 2026 awards — roadbuilders.bc.ca/tender-report
PUBLIC TENDER	CanadaBuys (federal procurement portal) — canadabuys.canada.ca
PUBLIC TENDER	SaskTenders — sasktenders.ca
PUBLIC TENDER	Manitoba Infrastructure — Tendering & Contracts — gov.mb.ca/mti/contracts/bidresults.html
PUBLIC TENDER	MERX — merx.com
MINING DATA	Natural Resources Canada — Mining Capital Expenditures
MINING DATA	SEDAR+ — NI 43-101 Technical Reports for comparable projects

25. REAL MARKET BENCHMARK — SME RANGES vs PUBLIC TENDER ACTUALS [Rev 2]

This section cross-references the SME unit-rate ranges in this manual against the highest-quality public-tender benchmark dataset available in Canada: Alberta Transportation's 2026 Unit Price Averages (UPA), which publishes the weighted average of the three lowest bids on highway/civil tenders awarded between May 1, 2024 and September 30, 2025. The Excel workbook (sheet '25 Real Market Benchmark') contains the full table with colour-coded variance flags; this section summarises the conclusions.

25.1 Top-Level Read

- Where AB UPA falls WITHIN the SME floor/ceiling — the SME range is well-calibrated to real Canadian public-tender market. Use confidently.
- Where AB UPA falls BELOW the SME floor — SME range is conservative because mining-site SME includes additional camp / LOA, remote indirects, mining-spec QA, and risk premium. SME floor justified for remote mining; consider lower bound for accessible drive-in projects.
- Where AB UPA falls ABOVE the SME ceiling — typically the AB UPA item has additional specialty scope (bridge-grade quarry rock, specialty steel pipe, deck overlay) — not directly comparable. Review scope inclusion.
- AB UPA is HIGHWAY tender data, not mining tender data. It is the closest available public benchmark for civil unit rates in Canada. NI 43-101 filings have aggregate capex but rarely line-item unit rates publicly. Detailed mining-tender unit rates are commercial-in-confidence.
- To adapt AB UPA to a mining-site context, add: camp / LOA if remote; regional adjustor if outside Alberta; mining-spec QC and indirects.

25.2 Cross-Reference Highlights (Selected)

Item	AB UPA 2026	SME Range (CAD)	Verdict
Common excavation balanced (G225)	\$8.20 / m³	\$24–\$32 / m³ (C-10-002)	AB lower — no haul. With 5-10 km haul \$0.44/m³·km → \$10–\$13. Mining indirects make up gap.
Excavation+haul (G248+G249)	\$15.22 + \$0.44/km	\$24–\$32 / m³	AB at 5 km haul = \$17.42 / m³ — WITHIN at floor
Borrow contractor-supplied (G236)	\$24.40 / m³	\$22–\$45 / m³ (C-40-002)	WITHIN — strong calibration
Granular base course (B282)	\$34.27 / t	\$105–\$135 / m³ (C-10-009)	At 2.0 t/m³ = \$68.54/m³ — SME higher (mining haul, OH)
CSP culvert 900 mm (D430)	\$720.58 / m	\$680–\$880 / m (C-50-003)	WITHIN
CSP culvert 1000 mm (D431)	\$824.24 / m	\$800–\$1,020 / m (C-50-004)	WITHIN
Heavy rock riprap Class 1 (F500)	\$313.41 / m³	\$65–\$100 / m³ (C-20-002)	AB much higher — bridge-spec sized rock. SME is bulk mining riprap. Both valid in domain.
Concrete Class HPC (F841)	\$2,692.30 / m³	\$2,200–\$3,500 / m³ (C-60-002)	WITHIN
Concrete Class C footings (F834)	\$1,951.01 / m³	\$2,200–\$3,500 / m³	AB just below SME floor — mining indirects + cold weather pours

Item	AB UPA 2026	SME Range (CAD)	Verdict
Asphalt Superpave (Q998)	$147.89 / t	$45–$62 / m² (C-30-005)	At 100mm × 2.3 t/m³ = $34/m². SME is for thinner 50mm — mining OH
Erosion silt fence (E435)	$15.16 / m	$18–$28 / m (C-20-008)	AB just below SME floor — CLOSE WITHIN
Non-woven geotextile (E452)	$2.82 / m²	$3–$6 / m² (C-20-007)	AB just below SME floor (~6%) — CLOSE WITHIN
Concrete curb (X320)	$170.32 / m	$185–$265 / m (B-04-003)	AB just below SME floor — CLOSE WITHIN
Pre-cast street light base (U122)	$3,480.95 / ea	$1,850–$3,500 / ea (EL-70-003)	WITHIN

25.3 Conclusion

The SME ranges in this manual are well-calibrated against real Canadian public-tender data, with reasonable upward bias to reflect mining-site indirects, remoteness, and risk premium relative to accessible highway projects. Where SME range and AB UPA disagree by more than 30%, the discrepancy is almost always explained by scope difference (e.g. bridge-grade vs mining-site bulk rip-rap) or by mining-site indirect / camp / haul-distance loading. Use the SME range as the working benchmark; use AB UPA as a sanity check to question whether a project-specific quote is contaminated by scope creep or by a misapplied indirect.

— END OF DOCUMENT —

RMM-CIVIL-CANADA-2026-MANUAL-REV3 | Rev 3 — End-to-End + Build-Up + Restored Items | Q2-2026